

Quantum Field Theory

Lecture Notes

Quantization, Fock space, propagators, scattering, and renormalization.



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Preface: What These Notes Are For

Remark

These notes are designed as a continuation of the Classical Field Theory I–II sequence. They do not try to re-teach variational calculus, Lorentz covariance, Noether currents, gauge redundancy, or the classical equations of fundamental fields from the beginning. Instead, they start where the classical notes stop: with fields as dynamical systems ready to be quantized.

Quantum Field Theory After Classical Field Theory

Classical field theory teaches us to describe nature by local fields, actions, symmetries, conservation laws, and equations of motion. Quantum field theory keeps this architecture, but changes the meaning of the basic objects. A classical field configuration is no longer the final description of a physical state. It becomes an ingredient in a quantum theory whose observable content is encoded in states, operators, correlation functions, transition amplitudes, and probabilities.

The guiding idea of the first part of these notes is simple: a free classical field decomposes into normal modes, and each normal mode behaves like a harmonic oscillator. Quantizing the field therefore means quantizing infinitely many oscillators at once. The excitations of these oscillators are interpreted as particles.

What Will Not Be Repeated

The notes assume familiarity with the following ideas from classical field theory:

- the action principle and Euler–Lagrange equations for fields,
- canonical momenta and Hamiltonian density,
- Lorentz covariance and relativistic notation,
- real and complex scalar fields,
- electromagnetic and Yang–Mills gauge fields at the classical level,
- Noether’s theorem and conserved currents,
- Green’s functions as solutions of linear differential equations.

Whenever one of these topics is used in a quantum calculation, the notes recall just enough of the classical structure to make the calculation readable. The aim is not repetition, but conversion: classical structures are translated into quantum structures.

What Is New in QFT

The genuinely new concepts are the vacuum, Fock space, field operators, propagators, Wick contractions, Feynman diagrams, scattering amplitudes, loop corrections, regularization, renormalization, and scale dependence. These ideas are not decorative additions to classical field theory. They are the mechanisms that make quantum field theory predictive.

A central conceptual warning will appear repeatedly: a Feynman diagram is not a literal picture of tiny particles following tiny trajectories. It is a compact bookkeeping device for terms in a perturbative expansion of correlation functions or scattering amplitudes.

How to Read These Notes

A useful rhythm for reading a QFT calculation is:

classical field $\longrightarrow$ mode expansion $\longrightarrow$ operators $\longrightarrow$ states $\longrightarrow$ correlators $\longrightarrow$ amplitudes.

At first this may look like a change of notation. It is not. Each arrow changes what is meant by a physical prediction. The purpose of the notes is to make those changes explicit and technically usable.

Chapter 1

From Classical Normal Modes to Quantum Oscillators

Remark

A free field is the continuum limit of a system with many coupled degrees of freedom. Its normal modes behave like independent harmonic oscillators. The first idea of quantum field theory is to quantize these oscillators and to interpret their excitations as field quanta.

1.1 The Purpose of the First Step

The transition from classical field theory to quantum field theory can look too large if one begins immediately with operator-valued distributions, Fock spaces, and relativistic commutators. This chapter takes a slower route. It explains why the harmonic oscillator algebra appears at the beginning of quantum field theory.

The guiding chain of ideas is

classical field $\longrightarrow$ normal modes $\longrightarrow$ classical oscillators $\longrightarrow$ quantum oscillators $\longrightarrow$ field quanta.

The goal is not yet to construct the full canonical field algebra. That will be done in the next chapter. Here we only identify the mechanical structure hidden inside a free field.

1.2 A Finite System of Coupled Oscillators

Consider a finite chain of N identical masses connected by springs. Let $q_j(t)$ be the displacement of the j -th mass from equilibrium. A typical quadratic Lagrangian has the form

$$L = \sum_{j=1}^N \frac{m}{2} \dot{q}_j^2 - \sum_{j=1}^N \frac{\kappa}{2} (q_{j+1} - q_j)^2,$$

with boundary conditions chosen so that the notation is meaningful. The coordinates q_j are coupled because the potential energy depends on differences between neighbouring displacements.

The normal-mode method replaces the original coordinates by new coordinates $Q_r(t)$,

$$q_j(t) = \sum_r Q_r(t) u_r(j),$$

where $u_r(j)$ are spatial mode functions. With an appropriate normalization, the Lagrangian becomes diagonal:

$$L = \sum_r \left(\frac{1}{2} \dot{Q}_r^2 - \frac{1}{2} \omega_r^2 Q_r^2 \right).$$

Thus the coupled system is equivalent to a collection of independent harmonic oscillators.

Example

The original variables q_j describe local displacements of the masses. The normal coordinates Q_r describe collective oscillation patterns of the whole chain. Quantization is easiest in the normal coordinates because the modes are independent.

1.3 The Field as a Continuum Limit

A field is the continuum analogue of a system with one degree of freedom at each spatial point. In a one-dimensional chain with lattice spacing a , the site position is $x_j = ja$, and one may think schematically of

$$q_j(t) \approx \phi(t, x_j).$$

As $a \rightarrow 0$, sums over sites become spatial integrals, and discrete normal modes become Fourier modes.

For a real scalar field in a finite box of volume V , periodic boundary conditions make the allowed momenta discrete. A schematic Fourier expansion is

$$\phi(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{x}}.$$

For a real field the complex coefficients are not all independent; one must impose the appropriate reality condition. Equivalently, one may use real sine and cosine modes. The important point is independent of this choice: after the free field is decomposed into normal modes, each independent mode behaves like a classical oscillator.

1.4 Each Free Mode Is a Harmonic Oscillator

The free real scalar field has Lagrangian

$$L = \int d^3x \left(\frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2 \right).$$

Its equation of motion is the Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

After the free field is diagonalized into normal modes, the Lagrangian takes the schematic oscillator form

$$L = \sum_{\mathbf{k}} \left(\frac{1}{2} \dot{q}_{\mathbf{k}}^2 - \frac{1}{2} \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

The canonical momentum of a mode is

$$p_{\mathbf{k}} = \dot{q}_{\mathbf{k}},$$

and the Hamiltonian is

$$H = \sum_{\mathbf{k}} H_{\mathbf{k}}, \quad H_{\mathbf{k}} = \frac{1}{2} p_{\mathbf{k}}^2 + \frac{1}{2} \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2.$$

A free scalar field is therefore a collection of independent harmonic oscillators, one for each independent normal mode.

Remark

Interactions destroy this simple diagonal form. Perturbative QFT begins by quantizing the free oscillators exactly and then treating interactions as corrections to the free theory.

1.5 Quantizing One Harmonic Oscillator

Before quantizing all field modes, recall the single harmonic oscillator. The classical variables q and p become operators $\hat{q}$ and $\hat{p}$ satisfying

$$[\hat{q}, \hat{p}] = i.$$

The Hamiltonian is

$$\hat{H} = \frac{1}{2}\hat{p}^2 + \frac{1}{2}\omega^2\hat{q}^2.$$

Introduce the annihilation and creation operators

$$\hat{a} = \sqrt{\frac{\omega}{2}}\hat{q} + \frac{i}{\sqrt{2\omega}}\hat{p}, \quad \hat{a}^\dagger = \sqrt{\frac{\omega}{2}}\hat{q} - \frac{i}{\sqrt{2\omega}}\hat{p}.$$

Then

$$[\hat{a}, \hat{a}^\dagger] = 1,$$

and

$$\hat{H} = \omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right).$$

The operator $\hat{a}^\dagger \hat{a}$ counts oscillator excitations.

1.6 Number States and the Oscillator Vacuum

The oscillator vacuum is defined by

$$\hat{a}|0\rangle = 0.$$

The excited states are

$$|n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}|0\rangle, \quad n = 0, 1, 2, \dots$$

If

$$\hat{N} = \hat{a}^\dagger \hat{a},$$

then

$$\hat{N}|n\rangle = n|n\rangle, \quad \hat{H}|n\rangle = \omega \left(n + \frac{1}{2} \right) |n\rangle.$$

The vacuum is not a state of zero energy. It has the zero-point energy $\omega/2$. This fact becomes more delicate for fields because a field has infinitely many oscillator modes.

1.7 From One Oscillator to Many Field Modes

For a free field, every independent mode is quantized as an oscillator. Thus one introduces a pair of operators for each mode,

$$\hat{a}_{\mathbf{k}}, \quad \hat{a}_{\mathbf{k}}^\dagger.$$

In a finite box their algebra is

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = \delta_{\mathbf{k}, \mathbf{p}}, \quad [\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \quad [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{p}}^\dagger] = 0.$$

The free-field vacuum is the state annihilated by all annihilation operators,

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \quad \text{for all } \mathbf{k}.$$

A one-quantum state is obtained by applying one creation operator:

$$|\mathbf{k}\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

A many-quantum state is obtained by applying several creation operators.

Definition

The *Fock space* of the free field is the Hilbert space generated from the vacuum by applying creation operators for all modes. It naturally contains states with zero, one, two, and more field quanta.

1.8 Field Quanta as Mode Excitations

For the free theory, the Hamiltonian has the schematic form

$$\hat{H} = \sum_{\mathbf{k}} \omega_{\mathbf{k}} \left(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2} \right).$$

The operator $\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}$ counts how many quanta occupy the mode $\mathbf{k}$. The state $\hat{a}_{\mathbf{k}}^\dagger|0\rangle$ has one quantum of energy $\omega_{\mathbf{k}}$ above the free vacuum and momentum $\mathbf{k}$. In the free theory, this is what one means by a one-particle state.

This interpretation is precise for a free field. In interacting QFT, particles can be dressed by interactions, unstable particles appear as resonances, and the vacuum itself can be nontrivial. Nevertheless, the free Fock-space picture is the starting point for perturbation theory and scattering.

1.9 What Is Still Missing?

The oscillator construction explains why creation and annihilation operators appear, but it is not yet the full canonical quantization of a field. We have not yet imposed commutation relations directly on $\hat{\phi}(t, \mathbf{x})$ and its conjugate momentum $\hat{\pi}(t, \mathbf{x})$. We have also not fixed the continuum normalization of the mode expansion, nor have we discussed fields as operator-valued distributions.

These are the tasks of the next chapter. Chapter 2 turns the oscillator picture into the canonical quantization of the real scalar field.

Summary

A free classical field can be diagonalized into independent normal modes. Each mode is a harmonic oscillator. Quantizing those oscillators gives creation and annihilation operators. Acting with creation operators on the vacuum builds Fock space, and the resulting mode excitations are interpreted as free particles. This oscillator picture is the conceptual bridge to canonical field quantization.

1.10 Chapter Checklist

After reading this chapter, one should be able to:

- explain why a free field decomposes into normal modes;
- explain why each independent mode behaves as a harmonic oscillator;
- quantize a single harmonic oscillator using $\hat{a}$ and $\hat{a}^\dagger$;
- define the oscillator vacuum and number states;
- describe the free-field vacuum and one-particle states;
- explain why free particles are mode excitations;
- state what remains to be done in canonical field quantization.

Chapter 2

Canonical Quantization of the Real Scalar Field

Remark

Chapter 1 explained why a free field should be thought of as a collection of oscillators. This chapter makes that statement precise. We impose canonical commutation relations on the field and its conjugate momentum, derive the standard mode expansion, and recover the Fock-space interpretation from the field algebra itself.

2.1 From the Oscillator Picture to Field Operators

The previous chapter started from normal modes. It argued that a free scalar field is a collection of harmonic oscillators labelled by momentum. That is the right intuition, but it is not yet the full canonical quantization of a field.

The full canonical construction begins before the normal-mode expansion. It starts with the classical field $\phi(t, \mathbf{x})$, its conjugate momentum $\pi(t, \mathbf{x})$, and their Poisson brackets. Quantization then promotes ϕ and π to operators and replaces the Poisson brackets by commutators.

The logic of this chapter is therefore

classical field variables $\longrightarrow$ equal-time commutators $\longrightarrow$ operator mode expansion $\longrightarrow$ creation and annihilation

This is the formal version of the oscillator picture developed in Chapter 1.

2.2 Classical Starting Point

We use units $c = \hbar = 1$ and the metric convention $(+, -, -, -)$. The free real scalar field is described by the Lagrangian density

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 = \frac{1}{2} \dot{\phi}^2 - \frac{1}{2} (\nabla \phi)^2 - \frac{1}{2} m^2 \phi^2.$$

The conjugate momentum is

$$\pi(t, \mathbf{x}) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}(t, \mathbf{x})} = \dot{\phi}(t, \mathbf{x}).$$

The Euler–Lagrange equation is the Klein–Gordon equation,

$$(\square + m^2)\phi(x) = 0,$$

where $x = (t, \mathbf{x})$. In components this reads

$$(\partial_t^2 - \nabla^2 + m^2)\phi(t, \mathbf{x}) = 0.$$

The classical Hamiltonian density is

$$\mathcal{H} = \frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2,$$

so the Hamiltonian is

$$H = \int d^3x \left(\frac{1}{2} \pi^2 + \frac{1}{2} (\nabla \phi)^2 + \frac{1}{2} m^2 \phi^2 \right).$$

This is the continuum field-theoretic analogue of the oscillator Hamiltonian $p^2/2 + \omega^2 q^2/2$.

2.3 Equal-Time Commutation Relations

The classical canonical Poisson bracket is

$$\{\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\}_{\text{PB}} = \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

Canonical quantization promotes ϕ and π to operators $\hat{\phi}$ and $\hat{\pi}$, and imposes

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

The remaining equal-time commutators vanish:

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0, \quad [\hat{\pi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = 0.$$

These relations are the field version of

$$[\hat{q}, \hat{p}] = i.$$

Instead of one coordinate and one momentum, there is one canonical pair at every point of space.

Definition

The *equal-time canonical commutation relations* are the basic operator relations between $\hat{\phi}$ and $\hat{\pi}$ at a fixed time. For bosonic fields they are commutators; for fermionic fields they will be replaced by anticommutators.

2.4 Finite Box and Continuum Normalization

Chapter 1 used a finite box to keep the momentum labels discrete. In a finite box, one writes sums over momenta and Kronecker deltas. In infinite volume, one uses integrals and Dirac deltas. The replacement rules are schematically

$$\frac{1}{V} \sum_{\mathbf{k}} \longrightarrow \int \frac{d^3k}{(2\pi)^3}, \quad V\delta_{\mathbf{k},\mathbf{p}} \longrightarrow (2\pi)^3\delta^{(3)}(\mathbf{k} - \mathbf{p}).$$

The continuum formulas below use the second convention. They look slightly more complicated than finite-box formulas, but they are the standard relativistic normalization used in scattering theory.

2.5 Mode Expansion of the Field Operator

The quantum field operator is expanded in positive- and negative-frequency solutions of the Klein–Gordon equation:

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

Since the field is real, $\hat{\phi}$ is Hermitian:

$$\hat{\phi}(t, \mathbf{x})^\dagger = \hat{\phi}(t, \mathbf{x}).$$

The appearance of both $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$ is the operator version of the reality condition on a classical real field.

The conjugate momentum operator is

$$\hat{\pi}(t, \mathbf{x}) = \partial_t \hat{\phi}(t, \mathbf{x}),$$

therefore

$$\hat{\pi}(t, \mathbf{x}) = -i \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_{\mathbf{k}}}{2}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} - \hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right).$$

The normalization factors are chosen so that the equal-time field commutators are satisfied.

2.6 Ladder-Operator Algebra

The canonical field commutators imply

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

with

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}] = 0, \quad [\hat{a}_{\mathbf{k}}^\dagger, \hat{a}_{\mathbf{p}}^\dagger] = 0.$$

Conversely, these ladder-operator relations imply the equal-time commutators of $\hat{\phi}$ and $\hat{\pi}$. This is the precise connection between the canonical field algebra and the oscillator algebra of Chapter 1.

The operator $\hat{a}_{\mathbf{k}}^\dagger$ creates a quantum in the momentum mode $\mathbf{k}$, while $\hat{a}_{\mathbf{k}}$ annihilates such a quantum. For the real scalar field there is only one kind of neutral scalar quantum.

2.7 Fock Space Revisited

The free-field vacuum is defined by

$$\hat{a}_{\mathbf{k}}|0\rangle = 0 \quad \text{for every } \mathbf{k}.$$

A one-particle momentum eigenstate is

$$|\mathbf{k}\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

A two-particle state is

$$|\mathbf{k}_1, \mathbf{k}_2\rangle = \hat{a}_{\mathbf{k}_1}^\dagger \hat{a}_{\mathbf{k}_2}^\dagger|0\rangle,$$

up to normalization. Since the creation operators commute, the scalar quanta are bosons.

A realistic one-particle state is usually a wave packet rather than a single momentum eigenstate:

$$|f\rangle = \int \frac{d^3k}{(2\pi)^3} f(\mathbf{k}) \hat{a}_{\mathbf{k}}^\dagger|0\rangle.$$

The function $f(\mathbf{k})$ describes the momentum profile of the state. This is the first place where the particle idea becomes more refined than the simple phrase “one quantum of one mode.”

2.8 Hamiltonian and Momentum Operators

The quantum Hamiltonian is obtained from the classical Hamiltonian by replacing ϕ and π by field operators:

$$\hat{H} = \int d^3x \left(\frac{1}{2} \hat{\pi}^2 + \frac{1}{2} (\nabla \hat{\phi})^2 + \frac{1}{2} m^2 \hat{\phi}^2 \right).$$

Substituting the mode expansion gives the formal expression

$$\hat{H} = \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \left(\hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + \frac{1}{2} (2\pi)^3 \delta^{(3)}(0) \right).$$

The first term counts excitations; the second term is the continuum notation for the sum of oscillator zero-point energies.

The spatial momentum operator is

$$\hat{\mathbf{P}} = \int d^3x \hat{\pi} \nabla \hat{\phi}.$$

With the same ordering convention, it becomes

$$\hat{\mathbf{P}} = \int \frac{d^3k}{(2\pi)^3} \mathbf{k} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}}.$$

Thus

$$\hat{H} \hat{a}_{\mathbf{k}}^\dagger |0\rangle = \omega_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger |0\rangle \quad \text{above the vacuum energy,}$$

and

$$\hat{\mathbf{P}} \hat{a}_{\mathbf{k}}^\dagger |0\rangle = \mathbf{k} \hat{a}_{\mathbf{k}}^\dagger |0\rangle.$$

The dispersion relation is

$$\omega_{\mathbf{k}}^2 = \mathbf{k}^2 + m^2.$$

This is why m is interpreted as the mass of the scalar quantum.

2.9 Normal Ordering

The vacuum-energy term in $\hat{H}$ is infinite in the continuum limit. For the free theory, it is common to define the normal-ordered Hamiltonian by moving all creation operators to the left of all annihilation operators:

$$: \hat{a}_{\mathbf{k}} \hat{a}_{\mathbf{p}}^\dagger := \hat{a}_{\mathbf{p}}^\dagger \hat{a}_{\mathbf{k}}.$$

Then

$$: \hat{H} := \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}} \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}},$$

and

$$\langle 0 | : \hat{H} : | 0 \rangle = 0.$$

Normal ordering removes the constant free vacuum energy from this Hamiltonian. It does not remove all divergences of interacting quantum field theory. It is a useful convention at the free-field level, not a substitute for renormalization.

2.10 Zero-Point Energy

In a finite box, the vacuum energy of the free scalar field is

$$E_0 = \frac{1}{2} \sum_{\mathbf{k}} \omega_{\mathbf{k}}.$$

In infinite-volume notation this becomes

$$E_0 = \frac{V}{2} \int \frac{d^3k}{(2\pi)^3} \omega_{\mathbf{k}}.$$

The integral diverges at large momentum. The divergence is a reminder that a field has infinitely many short-distance degrees of freedom.

For many non-gravitational scattering problems, only energy differences above the free vacuum matter, and normal ordering is sufficient. In contexts where vacuum energy gravitates, or where the vacuum itself changes, the issue becomes more subtle. The present course will return to this point when discussing renormalization and vacuum structure.

2.11 Fields as Operator-Valued Distributions

The notation $\hat{\phi}(x)$ suggests an operator at a point. This notation is useful, but local quantum fields are more accurately understood as operator-valued distributions. They become well-defined only after smearing with test functions.

At fixed time, a smeared field is

$$\hat{\phi}(f) = \int d^3x f(\mathbf{x}) \hat{\phi}(t, \mathbf{x}).$$

Products such as $\hat{\phi}(x)^2$ at the same spacetime point are generally singular and require a prescription. This is one of the structural reasons why renormalization appears in interacting field theory.

This point also clarifies the role of the mode expansion. The formula for $\hat{\phi}(t, \mathbf{x})$ is a compact way to encode how the field acts after it has been smeared. It should not be treated as an ordinary function with operator coefficients.

2.12 Microcausality

Relativistic quantum field theory must respect the causal structure of spacetime. For a scalar field, the basic local causality condition is

$$[\hat{\phi}(x), \hat{\phi}(y)] = 0 \quad \text{if} \quad (x - y)^2 < 0.$$

This condition is called *microcausality*. It says that field observables associated with spacelike separated regions commute.

For the free scalar field one can write

$$[\hat{\phi}(x), \hat{\phi}(y)] = i\Delta(x - y),$$

where Δ is the Pauli–Jordan commutator function. This function vanishes outside the light cone. The equal-time relation

$$[\hat{\phi}(t, \mathbf{x}), \hat{\phi}(t, \mathbf{y})] = 0$$

is therefore only the simplest special case of the full spacetime causality condition.

Summary

Canonical quantization begins with $\hat{\phi}$ and $\hat{\pi}$, not directly with particles. The equal-time commutation relations determine the ladder operator algebra in the mode expansion. The free-field vacuum and Fock space are then recovered from that algebra, matching the oscillator picture of Chapter 1. The new lessons are the precise continuum normalization, the Hamiltonian and momentum operators, normal ordering, the distributional nature of fields, and microcausality.

2.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the classical Lagrangian and Hamiltonian of the free real scalar field;
- derive the conjugate momentum $\pi = \dot{\phi}$;
- state the equal-time canonical commutation relations;

- write the mode expansions of $\hat{\phi}$ and $\hat{\pi}$;
- derive the ladder-operator commutators from the canonical field algebra;
- construct the vacuum, one-particle states, wave packets, and many-particle states;
- express $\hat{H}$ and $\hat{\mathbf{P}}$ in terms of ladder operators;
- explain what normal ordering removes and what it does not remove;
- explain why fields are operator-valued distributions;
- state the microcausality condition for a scalar field.

Chapter 3

The Complex Scalar Field and Antiparticles

Remark

The real scalar field has one neutral quantum for each momentum mode. The complex scalar field is the next step: the field is no longer Hermitian, and its quantization produces two independent kinds of quanta. They are naturally interpreted as particles and antiparticles.

3.1 From a Real Field to a Complex Field

Chapter 2 quantized a real scalar field. The reality condition $\hat{\phi}^\dagger = \hat{\phi}$ forced the same field operator to contain both annihilation and creation operators. A complex scalar field is different. The field ϕ and its Hermitian conjugate $\phi^\dagger$ are independent classical variables before quantization.

One may think of a complex scalar field as two real scalar fields packaged into one object:

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \quad \phi^\dagger = \frac{1}{\sqrt{2}}(\phi_1 - i\phi_2).$$

The important new feature is that the theory has a global phase symmetry. The field may be rotated by a constant phase,

$$\phi \longrightarrow e^{-i\alpha}\phi, \quad \phi^\dagger \longrightarrow e^{i\alpha}\phi^\dagger.$$

By Noether's theorem this symmetry gives a conserved charge. In the quantum theory, that charge will distinguish particles from antiparticles.

3.2 Classical Lagrangian and Canonical Momenta

The free complex scalar field is described by

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi^\dagger \phi.$$

In terms of time and space derivatives,

$$\mathcal{L} = \dot{\phi}^\dagger \dot{\phi} - \nabla \phi^\dagger \cdot \nabla \phi - m^2 \phi^\dagger \phi.$$

The conjugate momenta are

$$\pi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \dot{\phi}^\dagger, \quad \pi^\dagger = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger} = \dot{\phi}.$$

Thus π is conjugate to ϕ , while $\pi^\dagger$ is conjugate to $\phi^\dagger$.

The equations of motion are

$$(\square + m^2)\phi = 0, \quad (\square + m^2)\phi^\dagger = 0.$$

Both components satisfy the Klein–Gordon equation, but the quantum interpretation will be richer than in the real case.

3.3 Canonical Commutation Relations

Canonical quantization promotes $\phi, \phi^\dagger, \pi, \pi^\dagger$ to operators. The nonzero equal-time commutators are

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}),$$

and

$$[\hat{\phi}^\dagger(t, \mathbf{x}), \hat{\pi}^\dagger(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

All other equal-time commutators vanish. In particular,

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}^\dagger(t, \mathbf{y})] = 0, \quad [\hat{\phi}^\dagger(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = 0.$$

This is the canonical field algebra for a charged bosonic scalar field.

3.4 Mode Expansion

The complex field is not Hermitian, so its mode expansion cannot be the same as the real scalar expansion. The standard continuum expansion is

$$\hat{\phi}(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} + \hat{b}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} \right),$$

where

$$\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}.$$

The Hermitian conjugate field is

$$\hat{\phi}^\dagger(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{\mathbf{k}}}} \left(\hat{a}_{\mathbf{k}}^\dagger e^{i\omega_{\mathbf{k}}t - i\mathbf{k}\cdot\mathbf{x}} + \hat{b}_{\mathbf{k}} e^{-i\omega_{\mathbf{k}}t + i\mathbf{k}\cdot\mathbf{x}} \right).$$

There are now two independent sets of ladder operators: the a -operators and the b -operators. This is the algebraic origin of antiparticles in the complex scalar theory.

3.5 Two Independent Oscillator Families

The nonzero ladder-operator commutators are

$$[\hat{a}_{\mathbf{k}}, \hat{a}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}),$$

and

$$[\hat{b}_{\mathbf{k}}, \hat{b}_{\mathbf{p}}^\dagger] = (2\pi)^3 \delta^{(3)}(\mathbf{k} - \mathbf{p}).$$

The two families commute with each other:

$$[\hat{a}_{\mathbf{k}}, \hat{b}_{\mathbf{p}}] = 0, \quad [\hat{a}_{\mathbf{k}}, \hat{b}_{\mathbf{p}}^\dagger] = 0,$$

with similar relations for their Hermitian conjugates.

The vacuum is defined by

$$\hat{a}_{\mathbf{k}}|0\rangle = 0, \quad \hat{b}_{\mathbf{k}}|0\rangle = 0 \quad \text{for every } \mathbf{k}.$$

There are two basic one-particle states:

$$|\mathbf{k}; a\rangle = \hat{a}_{\mathbf{k}}^\dagger|0\rangle, \quad |\mathbf{k}; b\rangle = \hat{b}_{\mathbf{k}}^\dagger|0\rangle.$$

They have the same mass and the same energy-momentum relation, but opposite charges.

3.6 Hamiltonian and Momentum

After normal ordering, the Hamiltonian of the free complex scalar field is

$$:\hat{H}:=\int\frac{d^3k}{(2\pi)^3}\omega_{\mathbf{k}}\left(\hat{a}_{\mathbf{k}}^\dagger\hat{a}_{\mathbf{k}}+\hat{b}_{\mathbf{k}}^\dagger\hat{b}_{\mathbf{k}}\right).$$

The momentum operator is

$$:\hat{\mathbf{P}}:=\int\frac{d^3k}{(2\pi)^3}\mathbf{k}\left(\hat{a}_{\mathbf{k}}^\dagger\hat{a}_{\mathbf{k}}+\hat{b}_{\mathbf{k}}^\dagger\hat{b}_{\mathbf{k}}\right).$$

Both kinds of quanta carry positive energy. The difference between them is not energy or mass, but charge.

3.7 The Charge Operator

The global phase symmetry gives a conserved current. With the convention $\phi \rightarrow e^{-i\alpha}\phi$, the corresponding quantum charge may be written, after normal ordering, as

$$\hat{Q}=\int\frac{d^3k}{(2\pi)^3}\left(\hat{a}_{\mathbf{k}}^\dagger\hat{a}_{\mathbf{k}}-\hat{b}_{\mathbf{k}}^\dagger\hat{b}_{\mathbf{k}}\right).$$

This convention gives

$$\hat{Q}\hat{a}_{\mathbf{k}}^\dagger|0\rangle=+\hat{a}_{\mathbf{k}}^\dagger|0\rangle,\quad\hat{Q}\hat{b}_{\mathbf{k}}^\dagger|0\rangle=-\hat{b}_{\mathbf{k}}^\dagger|0\rangle.$$

Thus the a -quanta have charge $+1$, while the b -quanta have charge -1 . They are particle and antiparticle excitations of the same complex field.

The charge operator also generates the phase transformation of the field:

$$[\hat{Q},\hat{\phi}(x)]=-\hat{\phi}(x),\quad[\hat{Q},\hat{\phi}^\dagger(x)]=+\hat{\phi}^\dagger(x).$$

This is the operator statement that $\hat{\phi}$ carries charge -1 as an operator, even though $\hat{a}^\dagger$ creates a charge $+1$ state in the chosen convention.

3.8 Positive and Negative Frequency Revisited

In the real scalar field, the negative-frequency part was simply the Hermitian conjugate of the positive-frequency part of the same field. In the complex field, the negative-frequency part of $\hat{\phi}$ contains $\hat{b}^\dagger$, not $\hat{a}^\dagger$. This is the crucial difference.

The field $\hat{\phi}$ can annihilate an a -particle or create a b -antiparticle. The conjugate field $\hat{\phi}^\dagger$ can create an a -particle or annihilate a b -antiparticle. Symbolically,

$$\hat{\phi}\sim\hat{a}+\hat{b}^\dagger,\quad\hat{\phi}^\dagger\sim\hat{a}^\dagger+\hat{b}.$$

This is one of the cleanest ways to see why relativistic quantum fields contain antiparticles.

3.9 Why Antiparticles Appear Naturally

The appearance of antiparticles is not an optional addition. It follows from combining three requirements:

- relativistic positive-energy modes;
- canonical quantization;

- a complex field with a conserved charge.

The field must contain both positive- and negative-frequency solutions in order to be local and to satisfy the canonical commutation relations. For a charged field, those two frequency sectors are associated with two independent oscillator families. The quanta created by the second family have the opposite charge and are interpreted as antiparticles.

3.10 Complex Scalar Propagator

The natural two-point function of the complex scalar field is

$$D_F(x-y) = \langle 0 | T \hat{\phi}(x) \hat{\phi}^\dagger(y) | 0 \rangle.$$

It has the same momentum-space denominator as the real scalar propagator:

$$D_F(x-y) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik \cdot (x-y)}.$$

The difference is in the charge flow. The propagator connects ϕ with $\phi^\dagger$, because charge must be conserved. Correlators such as $\langle 0 | T \hat{\phi}(x) \hat{\phi}(y) | 0 \rangle$ vanish in the free theory when the vacuum has zero charge.

3.11 Physical Interpretation of the Fock Space

The Fock space of the complex scalar field contains particles, antiparticles, and arbitrary multiparticle states. Examples are

$$\hat{a}_{\mathbf{k}}^\dagger |0\rangle, \quad \hat{b}_{\mathbf{p}}^\dagger |0\rangle, \quad \hat{a}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{p}}^\dagger |0\rangle.$$

The last state contains one particle and one antiparticle. Its total charge is zero, but it is not the vacuum. Charge conservation restricts which transitions are possible once interactions are introduced.

Summary

A complex scalar field is equivalent to two real scalar degrees of freedom, but its global phase symmetry gives a conserved charge. Quantization produces two independent oscillator families. One creates particles and the other creates antiparticles, with equal mass and opposite charge. The field $\hat{\phi}$ contains an annihilation operator for particles and a creation operator for antiparticles, while $\hat{\phi}^\dagger$ does the reverse.

3.12 Chapter Checklist

After reading this chapter, one should be able to:

- explain why ϕ and $\phi^\dagger$ are independent field variables before quantization;
- write the canonical momenta of the complex scalar field;
- state the equal-time commutation relations;
- write the mode expansions of $\hat{\phi}$ and $\hat{\phi}^\dagger$;
- explain why two independent ladder-operator families appear;

- construct particle and antiparticle states;
- write the charge operator and interpret its eigenvalues;
- explain why antiparticles are required in a relativistic charged field;
- identify the charge structure of the complex scalar propagator.

Chapter 4

Quantization of the Dirac Field

Remark

Scalar fields describe spin-zero quanta and are quantized with commutators. The Dirac field describes spin-1/2 quanta. Its consistent quantization requires anticommutators, and its mode expansion naturally contains particle and antiparticle operators with spin labels.

4.1 Why Fermionic Quantization Is Different

Chapters 2 and 3 quantized bosonic scalar fields. The ladder operators obeyed commutation relations, and many-particle states were symmetric under exchange. The Dirac field is different because its quanta are fermions. Fermionic states must be antisymmetric under exchange, and no two identical fermions can occupy the same one-particle state.

This difference is implemented algebraically by replacing commutators with anticommutators. For operators $\hat{A}$ and $\hat{B}$, the anticommutator is

$$\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}.$$

The quantized Dirac field will be built from creation and annihilation operators whose anticommutation relations enforce Fermi statistics.

There is also a second important difference. The Dirac equation is first order in time derivatives, unlike the Klein–Gordon equation. This changes the canonical structure and makes the Dirac field look less like a collection of ordinary oscillator coordinates. Nevertheless, the final Fock-space structure is parallel to the scalar case: there are annihilation operators, creation operators, a vacuum, particles, antiparticles, and number operators.

4.2 Classical Dirac Field

The Dirac field is a spinor field $\psi(x)$. Its Dirac adjoint is

$$\bar{\psi} = \psi^\dagger \gamma^0.$$

The free Dirac Lagrangian density is

$$\mathcal{L} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi.$$

The equation of motion is the Dirac equation,

$$(i\gamma^\mu \partial_\mu - m)\psi(x) = 0.$$

The adjoint field satisfies

$$\bar{\psi}(x)(i\overleftarrow{\partial}_\mu \gamma^\mu + m) = 0,$$

where the arrow indicates that the derivative acts to the left.

The matrices γ^μ obey the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}.$$

This algebra is what makes the Dirac equation compatible with the relativistic mass-shell relation $p^2 = m^2$.

4.3 Canonical Structure and Anticommutation Relations

Because the Lagrangian is first order in $\dot{\psi}$, the momentum conjugate to ψ is proportional to $\psi^\dagger$:

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\psi^\dagger.$$

The quantum theory is defined by equal-time anticommutation relations,

$$\{\hat{\psi}_\alpha(t, \mathbf{x}), \hat{\psi}_\beta^\dagger(t, \mathbf{y})\} = \delta_{\alpha\beta} \delta^{(3)}(\mathbf{x} - \mathbf{y}),$$

with

$$\{\hat{\psi}_\alpha(t, \mathbf{x}), \hat{\psi}_\beta(t, \mathbf{y})\} = 0, \quad \{\hat{\psi}_\alpha^\dagger(t, \mathbf{x}), \hat{\psi}_\beta^\dagger(t, \mathbf{y})\} = 0.$$

The spinor indices α, β label the components of the Dirac spinor.

Remark

The replacement of commutators by anticommutators is not a cosmetic change. It is what produces fermionic Fock space, positive-energy particle interpretation, and the Pauli exclusion principle.

4.4 Mode Expansion of the Dirac Field

The free Dirac equation has positive-energy spinor solutions $u_s(\mathbf{p})$ and negative-frequency spinor solutions $v_s(\mathbf{p})$, where s labels spin. A standard expansion is

$$\hat{\psi}(x) = \sum_s \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(\hat{b}_s(\mathbf{p}) u_s(\mathbf{p}) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} + \hat{d}_s^\dagger(\mathbf{p}) v_s(\mathbf{p}) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}} \right),$$

where

$$E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}.$$

The adjoint field is

$$\hat{\bar{\psi}}(x) = \sum_s \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}}}} \left(\hat{b}_s^\dagger(\mathbf{p}) \bar{u}_s(\mathbf{p}) e^{iE_{\mathbf{p}}t - i\mathbf{p}\cdot\mathbf{x}} + \hat{d}_s(\mathbf{p}) \bar{v}_s(\mathbf{p}) e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}} \right).$$

The structure is analogous to the complex scalar field:

$$\hat{\psi} \sim \hat{b} + \hat{d}^\dagger, \quad \hat{\bar{\psi}} \sim \hat{b}^\dagger + \hat{d}.$$

The operators $\hat{b}^\dagger$ create particles, while $\hat{d}^\dagger$ create antiparticles.

4.5 Spinor Creation and Annihilation Operators

The nonzero anticommutators are

$$\{\hat{b}_s(\mathbf{p}), \hat{b}_r^\dagger(\mathbf{q})\} = (2\pi)^3 \delta_{sr} \delta^{(3)}(\mathbf{p} - \mathbf{q}),$$

and

$$\{\hat{d}_s(\mathbf{p}), \hat{d}_r^\dagger(\mathbf{q})\} = (2\pi)^3 \delta_{sr} \delta^{(3)}(\mathbf{p} - \mathbf{q}).$$

All other anticommutators vanish. In particular,

$$\{\hat{b}_s(\mathbf{p}), \hat{d}_r(\mathbf{q})\} = 0, \quad \{\hat{b}_s(\mathbf{p}), \hat{d}_r^\dagger(\mathbf{q})\} = 0.$$

The label s is the spin label. For a massive Dirac particle there are two spin states for the particle and two spin states for the antiparticle.

Because the creation operators anticommute, applying the same creation operator twice gives zero:

$$\left(\hat{b}_s^\dagger(\mathbf{p})\right)^2 = 0, \quad \left(\hat{d}_s^\dagger(\mathbf{p})\right)^2 = 0.$$

This is the algebraic form of the Pauli exclusion principle.

4.6 Particle and Antiparticle States

The vacuum is defined by

$$\hat{b}_s(\mathbf{p})|0\rangle = 0, \quad \hat{d}_s(\mathbf{p})|0\rangle = 0 \quad \text{for all } \mathbf{p}, s.$$

A one-particle state is

$$|\mathbf{p}, s; b\rangle = \hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

A one-antiparticle state is

$$|\mathbf{p}, s; d\rangle = \hat{d}_s^\dagger(\mathbf{p})|0\rangle.$$

For the electron field, the $b^\dagger$ -states are electron states and the $d^\dagger$ -states are positron states. More generally, a Dirac field creates and annihilates a charged fermion and its antiparticle.

A two-fermion state changes sign when the order of the creation operators is exchanged:

$$\hat{b}_s^\dagger(\mathbf{p})\hat{b}_r^\dagger(\mathbf{q})|0\rangle = -\hat{b}_r^\dagger(\mathbf{q})\hat{b}_s^\dagger(\mathbf{p})|0\rangle.$$

This antisymmetry is the defining feature of fermionic Fock space.

4.7 Hamiltonian, Momentum, and Charge

After normal ordering, the free Dirac Hamiltonian is

$$:\hat{H}:= \sum_s \int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p}) \right).$$

The momentum operator is

$$:\hat{\mathbf{P}}:= \sum_s \int \frac{d^3p}{(2\pi)^3} \mathbf{p} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) + \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p}) \right).$$

Both particles and antiparticles carry positive energy.

For a Dirac field with unit charge convention, the normal-ordered charge has the form

$$\hat{Q} = \sum_s \int \frac{d^3p}{(2\pi)^3} \left(\hat{b}_s^\dagger(\mathbf{p})\hat{b}_s(\mathbf{p}) - \hat{d}_s^\dagger(\mathbf{p})\hat{d}_s(\mathbf{p}) \right).$$

For the electron field one often multiplies this operator by $-e$, so that particles created by $\hat{b}^\dagger$ have charge $-e$, and antiparticles created by $\hat{d}^\dagger$ have charge $+e$.

4.8 Why Anticommutators Give the Correct Energy Spectrum

The negative-frequency part of the Dirac field is not interpreted as a tower of negative-energy particles. After quantization, it is reinterpreted as the creation of antiparticles with positive energy. The anticommutation relations are essential for this reinterpretation to give a Hamiltonian bounded below.

Schematically, the Dirac Hamiltonian before normal ordering contains terms of the form

$$\hat{b}^\dagger \hat{b} - \hat{d} \hat{d}^\dagger.$$

Using anticommutators,

$$\hat{d} \hat{d}^\dagger = 1 - \hat{d}^\dagger \hat{d},$$

so, up to an infinite constant that is removed by normal ordering, the antiparticle contribution becomes positive:

$$\hat{d}^\dagger \hat{d}.$$

This is the algebraic reason the free Dirac Hamiltonian counts both particles and antiparticles with positive energy.

4.9 Spinor Sums and Completeness

The spinor wave functions are normalized so that they satisfy completeness relations. A common convention is

$$\sum_s u_s(\mathbf{p}) \bar{u}_s(\mathbf{p}) = \gamma^\mu p_\mu + m,$$

and

$$\sum_s v_s(\mathbf{p}) \bar{v}_s(\mathbf{p}) = \gamma^\mu p_\mu - m,$$

where $p^0 = E_{\mathbf{p}}$. These identities are used constantly when computing fermion propagators and scattering amplitudes.

The spinor sums play for the Dirac field a role analogous to polarization sums for vector fields. They allow spin labels on internal lines to be summed in a Lorentz-covariant way.

4.10 The Dirac Propagator

The Feynman propagator of the Dirac field is

$$S_F(x - y) = \langle 0 | T \hat{\psi}(x) \hat{\bar{\psi}}(y) | 0 \rangle.$$

In momentum space it is

$$S_F(x - y) = \int \frac{d^4 p}{(2\pi)^4} \frac{i(\gamma^\mu p_\mu + m)}{p^2 - m^2 + i\epsilon} e^{-ip \cdot (x - y)}.$$

Equivalently, the propagator is the Green function of the Dirac operator:

$$(i\gamma^\mu \partial_\mu - m) S_F(x - y) = i\delta^{(4)}(x - y).$$

Compared with the scalar propagator, the denominator is the same mass-shell factor, but the numerator carries spinor structure.

4.11 Chirality, Helicity, and the Massless Limit

The Dirac field contains spin degrees of freedom. For massive fermions, the spin label may be chosen in several equivalent ways. For massless fermions, helicity and chirality become especially important.

The chirality matrix γ^5 defines projection operators

$$P_L = \frac{1}{2}(1 - \gamma^5), \quad P_R = \frac{1}{2}(1 + \gamma^5).$$

They split a Dirac field into left- and right-chiral components:

$$\psi_L = P_L \psi, \quad \psi_R = P_R \psi.$$

For a massless Dirac field, the left- and right-chiral parts decouple. This will be essential later in the electroweak theory, where left- and right-handed fermions transform differently under gauge symmetries.

4.12 Majorana Fields: Conceptual Overview

A Dirac field describes a charged fermion and a distinct antiparticle. A Majorana field is different: it is a fermionic field that is equal to its charge conjugate in an appropriate sense. Informally, a Majorana fermion is its own antiparticle.

The Majorana idea is possible for neutral fermions, not for fields carrying an ordinary conserved electric charge. The full construction requires charge conjugation matrices and reality conditions on spinors. For now, the important contrast is simple:

Dirac field: particle and antiparticle are distinct,

while

Majorana field: particle and antiparticle are identified.

Majorana fields will become relevant when discussing neutrino masses and extensions of the Standard Model.

Summary

The Dirac field is the first fermionic quantum field in the course. Its quantization uses anticommutators rather than commutators. The mode expansion contains particle annihilation operators and antiparticle creation operators, now with spin labels. Anticommutation gives fermionic Fock space, the Pauli exclusion principle, and a Hamiltonian with positive particle and antiparticle energies after normal ordering.

4.13 Chapter Checklist

After reading this chapter, one should be able to:

- write the free Dirac Lagrangian and Dirac equation;
- explain why fermionic fields are quantized with anticommutators;
- write the mode expansion of $\hat{\psi}$ and $\hat{\bar{\psi}}$;
- state the anticommutation relations of $\hat{b}, \hat{b}^\dagger$ and $\hat{d}, \hat{d}^\dagger$;
- construct fermion and antifermion states;
- explain how the Hamiltonian counts positive-energy particles and antiparticles;
- write the charge operator and interpret the sign convention;
- state the standard spinor completeness relations;
- write the Dirac propagator;
- distinguish Dirac and Majorana fermions at the conceptual level.

Chapter 5

Quantization of the Electromagnetic Field

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

5.1 Planned Sections

- Why gauge fields are harder to quantize.
- Physical and redundant degrees of freedom.
- Coulomb gauge quantization.
- Photon creation and annihilation operators.
- Polarization vectors.
- Photon states and helicity.
- Covariant quantization.
- Gauge fixing and the photon propagator.
- Unphysical polarizations and their removal.
- Gauge-invariant observables.
- What is a photon?

Chapter 6

Propagators as Quantum Correlation Functions

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

6.1 Planned Sections

- From classical Green's functions to quantum two-point functions.
- Wightman functions.
- Time-ordered products.
- Feynman propagator.
- Retarded and advanced propagators in QFT.
- The $i\epsilon$ prescription.
- Momentum-space propagators.
- Pole structure and particle interpretation.
- Propagation vs correlation.
- What a propagator actually means.

Chapter 7

The Interaction Picture

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

7.1 Planned Sections

- Schrödinger, Heisenberg, and interaction pictures.
- Free fields as the starting point.
- Interaction Hamiltonian.
- Time evolution operator.
- Dyson series.
- Time-ordered exponential.
- Perturbation theory as an expansion in couplings.
- Vacuum persistence amplitude.
- Adiabatic switching and the meaning of in/out states.
- Why exact interacting QFT is difficult.

Chapter 8

Wick Theorem and Diagrammatics

Remark

This chapter follows the planned QFT table of contents. It is a placeholder to be developed in small increments.

8.1 Planned Sections

- Normal ordering revisited.
- Contractions.
- Wick theorem for scalar fields.
- Wick theorem for fermions.
- Fermionic sign factors.
- From algebra to diagrams.
- External lines, internal lines, and vertices.
- Connected and disconnected diagrams.
- Vacuum bubbles.
- Symmetry factors.
- Interpretation limits of Feynman diagrams.

Chapter 9

Scattering Theory and the S -Matrix

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

9.1 Planned Sections

- In-states and out-states.
- The S -matrix.
- Transition amplitudes.
- Matrix elements and momentum conservation.
- Cross sections.
- Decay rates.
- Lorentz-invariant phase space.
- The LSZ reduction formula.
- From correlation functions to scattering amplitudes.
- Tree-level $2 \rightarrow 2$ scattering.
- Decay of an unstable particle.
- From amplitudes to measurable predictions.

Chapter 10

Path Integral for Quantum Mechanics

Remark

This chapter is reserved by the course plan in `ideas_Quantum_Field_Theory.md`. It will be developed in small increments.

10.1 Planned Sections

- Transition amplitudes as sums over histories.
- Time slicing.
- Classical limit and stationary phase.
- Gaussian integrals.
- Generating functionals in ordinary quantum mechanics.
- Sources and correlation functions.
- Why the path integral is useful for QFT.
- Relation to canonical quantization.

Chapter 11

Path Integral for Scalar Fields

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

11.1 Planned Sections

- Functional integration.
- Free scalar generating functional.
- Sources and derivatives.
- Correlation functions.
- Gaussian functional integrals.
- Perturbation theory.
- Diagrammatic expansion.
- Connected functional.
- Effective action.
- Classical field as expectation value.
- One-particle-irreducible functions.

Chapter 12

Fermionic Path Integrals

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

12.1 Planned Sections

- Grassmann variables.
- Berezin integration.
- Gaussian integrals for Grassmann variables.
- Fermionic generating functional.
- Dirac propagator from the path integral.
- Fermion loops and minus signs.
- Determinants from integrating out fermions.
- Majorana fermions in the path integral.
- Physical interpretation and common pitfalls.

Chapter 13

Gauge Fields in the Path Integral

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

13.1 Planned Sections

- Overcounting gauge-equivalent configurations.
- Gauge fixing in the functional integral.
- Faddeev–Popov method.
- Ghost fields.
- Why ghosts are not physical particles.
- Gauge-fixed photon path integral.
- Gauge-fixed Yang–Mills path integral.
- First introduction to BRST symmetry.
- Gauge independence of physical observables.
- Meaning of gauge fixing in quantum theory.

Chapter 14

ϕ^4 Theory as the First Interacting QFT

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

14.1 Planned Role

This chapter will introduce the first interacting scalar quantum field theory and the basic structure of perturbative Feynman rules, loop corrections, counterterms, and renormalized parameters.

Chapter 15

Yukawa Theory

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

15.1 Planned Role

This chapter will study a scalar field coupled to fermions, Yukawa vertices, simple scattering and decay processes, and the low-energy interpretation of scalar exchange.

Chapter 16

Quantum Electrodynamics

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

16.1 Planned Role

This chapter will introduce QED as the quantum theory of charged matter and light, including Feynman rules, spinor chains, Ward identities, and standard tree-level processes.

Chapter 17

Loop Effects in QED

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

17.1 Planned Role

This chapter will cover radiative corrections in QED, including self-energies, vacuum polarization, vertex corrections, charge renormalization, infrared divergences, and inclusive observables.

Chapter 18

Regularization

Remark

Placeholder chapter from the QFT course plan in `ideas_Quantum_Field_Theory.md`.
Detailed lecture text will be added later.

18.1 Planned Role

This chapter will explain why divergent loop integrals appear and how cutoff, Pauli–Villars, dimensional regularization, and subtraction schemes organize intermediate infinities.

Chapter 19

Renormalization

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

19.1 Planned Role

This chapter will develop the standard QFT treatment of renormalized masses, fields, couplings, counterterms, and physical predictions.

Chapter 20

Renormalization Group

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

20.1 Planned Role

This chapter will organize the scale dependence of quantum field theories and introduce running couplings, beta functions, fixed points, and effective descriptions.

Chapter 21

Effective Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

21.1 Planned Role

This chapter will introduce effective field theory, separation of scales, matching, power counting, symmetries, operator bases, decoupling, and naturalness.

Chapter 22

Non-Abelian Gauge Theory as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

22.1 Planned Role

This chapter will develop quantum non-Abelian gauge fields, self-interactions, gauge-fixed Feynman rules, ghosts, Ward-type identities, and unitarity checks.

Chapter 23

BRST Symmetry

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

23.1 Planned Role

This chapter will introduce BRST transformations, ghost number, nilpotency, physical states as cohomology, gauge independence, and Slavnov–Taylor identities.

Chapter 24

Quantum Chromodynamics

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

24.1 Planned Role

This chapter will introduce QCD field content, color charge, QCD Feynman rules, gluon self-interactions, asymptotic freedom, confinement, jets, and perturbative versus nonperturbative regimes.

Chapter 25

Symmetries in Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

25.1 Planned Role

This chapter will translate classical symmetries into quantum operators, conserved charges, Ward identities, current correlation functions, selection rules, and constraints on observables.

Chapter 26

Anomalies

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

26.1 Planned Role

This chapter will study symmetries that fail after quantization, chiral currents, triangle diagrams, path-integral measure effects, gauge anomalies, anomaly cancellation, and physical consequences.

Chapter 27

Vacuum Structure and Nonperturbative Effects

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

27.1 Planned Role

This chapter will discuss the quantum vacuum, degenerate vacua, tunneling, instantons, theta vacua, false vacuum decay, semiclassical approximation, and nonperturbative contributions.

Chapter 28

The Quantum Electroweak Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

28.1 Planned Role

This chapter will introduce the quantum electroweak sector, gauge fixing after symmetry breaking, massive vector boson propagators, currents, weak processes, and the low-energy Fermi theory.

Chapter 29

The Standard Model Lagrangian as a Quantum Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

29.1 Planned Role

This chapter will organize the Standard Model field content, representations, gauge interactions, Yukawa sector, Higgs sector, flavor mixing, neutrino masses, anomaly cancellation, and EFT viewpoint.

Chapter 30

Selected Standard Model Processes

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

30.1 Planned Role

This chapter will connect QFT calculations with representative Standard Model predictions and collider observables.

Chapter 31

Euclidean Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

31.1 Planned Role

This chapter will introduce Wick rotation, Euclidean correlation functions, reflection positivity, statistical field theory connections, finite temperature, boundary conditions, and Matsubara frequencies.

Chapter 32

Lattice Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

32.1 Planned Role

This chapter will introduce lattice regularization, scalar fields on a lattice, gauge fields on links, Wilson action, Wilson loops, fermion difficulties, Monte Carlo sampling, and the continuum limit.

Chapter 33

Large- N and Semiclassical Methods

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

33.1 Planned Role

This chapter will introduce large- N counting, planar diagrams, saddle-point methods, semiclassical expansion, collective fields, solitons, instanton calculus, and reliability of approximations.

Chapter 34

Conceptual Synthesis I

34.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 35

Conceptual Synthesis II

35.1 Planned Role

Placeholder chapter from the QFT course plan.

Chapter 36

Summary: The Architecture of Quantum Field Theory

Remark

Placeholder chapter from the QFT course plan. Detailed lecture text will be added later.

36.1 Planned Role

This chapter will summarize quantization, Fock space, propagators, interactions, renormalization, gauge theory, anomalies, nonperturbative effects, the Standard Model, and open directions.

Appendix A

Quantization Dictionary

Remark

This appendix records the translation rules that are used repeatedly in the main text. They should not be treated as mechanical replacements valid in every context. They are starting points that must be checked against constraints, symmetries, domains of operators, and gauge redundancies.

A.1 Classical Mode to Quantum Oscillator

A classical normal mode with amplitude $q_{\mathbf{k}}(t)$ and frequency $\omega_{\mathbf{k}}$ is quantized like a harmonic oscillator. Schematically,

$$q_{\mathbf{k}}, p_{\mathbf{k}} \longrightarrow \hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}}, \quad [\hat{q}_{\mathbf{k}}, \hat{p}_{\mathbf{k}'}] = i\delta_{\mathbf{k}\mathbf{k}'}.$$

The oscillator is then rewritten in terms of ladder operators $\hat{a}_{\mathbf{k}}$ and $\hat{a}_{\mathbf{k}}^\dagger$.

A.2 Poisson Bracket to Commutator

The canonical rule is

$$\{A, B\}_{\text{PB}} \longrightarrow \frac{1}{i}[\hat{A}, \hat{B}].$$

For fields, the equal-time canonical relation is

$$[\hat{\phi}(t, \mathbf{x}), \hat{\pi}(t, \mathbf{y})] = i\delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

A.3 Classical Field to Field Operator

A classical field $\phi(x)$ becomes an operator-valued distribution $\hat{\phi}(x)$. The word “distribution” is important: products of field operators at the same spacetime point are generally singular and require a careful definition.

A.4 Classical Green’s Function to Propagator

A classical Green’s function solves a differential equation with a delta-function source. A quantum propagator is a vacuum expectation value of a time-ordered product,

$$\Delta_F(x - y) = \langle 0|T\{\hat{\phi}(x)\hat{\phi}(y)\}|0\rangle.$$

It is related to Green’s functions, but its physical meaning is quantum correlation, not merely classical signal propagation.

A.5 Classical Symmetry to Quantum Ward Identity

A classical continuous symmetry gives a conserved current. In the quantum theory, the same symmetry constrains correlation functions and scattering amplitudes through Ward identities. If the symmetry fails after quantization, the failure is called an anomaly.

Appendix B

Reference B

Placeholder appendix.

Appendix C

Reference C

Placeholder appendix.

Appendix D

Reference D

Placeholder appendix.

Appendix E

Reference E

Placeholder appendix.

List of Figures